

Library API

REST API Documentation

Version 1.0 • Laravel 11 • Sanctum Authentication
Base URL: http://library-api.test/api
June 2026

Framework	Laravel 11
Authentication	Laravel Sanctum (Bearer Token)
Format	JSON — Content-Type: application/json
Database	MySQL
Test Suite	43 tests, 119 assertions — all passing
User Types	Staff (admin / librarian) and Member

Authentication

All protected endpoints require an Authorization header:

Authorization: Bearer YOUR_TOKEN

Staff tokens are obtained via POST /api/staff/login

Member tokens are obtained via POST /api/member/login

Auth-Staff

POST

StaffLogin

Public

<http://library-api.test/api/staff/login>

No authentication required — public endpoint

REQUEST BODY

Staff credentials

```
{"email": "admin@library.com", "password": "password"}
```

EXAMPLE RESPONSE

```
{
  "message": "Login successful.",
  "token": "2|abc123...",
  "data": {
    "id": 1,
    "name": "Admin User",
    "email": "admin@library.com",
    "role": "admin"
  }
}
```

GET

Get Staff Profile

Bearer Token

<http://library-api.test/api/staff/me>

Staff token required (librarian or admin)

POST

Staff Logout

Bearer Token

<http://library-api.test/api/staff/logout>

Staff token required (librarian or admin)

Auth-Member

POST

MemberLogin

Public

<http://library-api.test/api/member/login>

No authentication required — public endpoint

REQUEST BODY

Member credentials

```
{"email": "ade@member.com", "password": "password"}
```

EXAMPLE RESPONSE

```
{
  "message": "Login successful.",
  "token": "5|xyz789...",
  "data": {
    "id": 1,
    "first_name": "Ade",
    "last_name": "Regular",
    "email": "ade@member.com"
  }
}
```

GET

Get Member Profile

Bearer Token

http://library-api.test/api/member/me

Member token required

POST

Member Logout

Bearer Token

http://library-api.test/api/member/logout

Member token required

Catalog - Categories

GET

List Categories

Public

http://library-api.test/api/categories

No authentication required — public endpoint

POST

Create Categories

Bearer Token

http://library-api.test/api/categories

Staff token required (librarian or admin)

REQUEST BODY

```
{"name": "Biography", "description": "Biographical works about real people"}
```

Catalog - Authors

GET

List Authors

Public

http://library-api.test/api/authors

No authentication required — public endpoint

POST

Create Authors

Bearer Token

http://library-api.test/api/authors

Staff token required (librarian or admin)

REQUEST BODY

```
{"name": "Ben Okri", "bio": "Nigerian poet and novelist, winner of the Booker Prize."}
```

PATCH

Update Authors

Bearer Token

http://library-api.test/api/authors/1

Staff token required (librarian or admin)

REQUEST BODY

```
{"bio": "Nigerian novelist, poet, and professor. Author of Things Fall Apart."}
```

Catalog - Books

GET

List Books

Public

http://library-api.test/api/books

No authentication required — public endpoint

GET

Filter Books by Title

Public

http://library-api.test/api/books?title=Things

No authentication required — public endpoint

GET

Get Single Book

Public

http://library-api.test/api/books/1

No authentication required — public endpoint

POST

Create Book

Bearer Token

http://library-api.test/api/books

Staff token required (librarian or admin)

REQUEST BODY

```
{
  "title": "Arrow of God",
  "isbn": "978-0-385-01480-0",
  "description": "Third novel in Chinua Achebe's African Trilogy.",
  "published_year": 1964,
  "author_ids": [
    1
  ],
  "category_ids": [
    1
  ],
  "copies": 2
}
```

PATCH

Update Book

Bearer Token

http://library-api.test/api/books/1

Staff token required (librarian or admin)

REQUEST BODY

```
{
  "description": "Chinua Achebe's debut novel, widely regarded as the finest African novel ever written."
}
```

DELETE

Retire Book

Bearer Token

<http://library-api.test/api/books/3>
Staff token required (librarian or admin)

BUSINESS RULES

- Cannot retire a book with active loans — returns 409
- Retired books are hidden from the public catalog

ERROR RESPONSES

Status	Message
409	Book has active loans and cannot be retired.

GET

Filter Book by Author

Public

<http://library-api.test/api/books?author=chinua>
No authentication required — public endpoint

GET

Filter Book By Category

Public

<http://library-api.test/api/books?category=Fiction>
No authentication required — public endpoint

Loans

POST

Borrow Book

Bearer Token

http://library-api.test/api/loans

Member token required

REQUEST BODY

Member must be active with no unpaid fines

```
{
  "book_id": 1
}
```

EXAMPLE RESPONSE

```
{
  "message": "Book borrowed successfully.",
  "data": {
    "id": 1,
    "status": "active",
    "borrowed_at": "2026-06-22 14:50:54",
    "due_date": "2026-07-06 14:50:54",
    "returned_at": null,
    "renewals_count": 0,
    "fines_accrued": "0.00",
    "is_overdue": false,
    "book": {
      "title": "Things Fall Apart",
      "isbn": "978-0-385-47454-2"
    },
    "member": {
      "first_name": "Ade",
      "last_name": "Regular",
      "email": "ade@member.com"
    }
  }
}
```

BUSINESS RULES

- Member must be active (not suspended)
- Member must have no unpaid fines
- Member cannot exceed their tier max_books limit
- Member cannot borrow the same book twice simultaneously
- A non-damaged, available copy must exist

ERROR RESPONSES

Status	Message
403	Member account is suspended.
403	Member has unpaid fines.
422	Member has reached their borrowing limit.
422	Member already has an active loan for this book.
422	No available copy found for this book.

GET

List Loans

Bearer Token

http://library-api.test/api/loans/1

Member token required

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Renew Loan

Bearer Token

<http://library-api.test/api/loans/1/renew>

Member token required

BUSINESS RULES

- Maximum 2 renewals per loan (MAX_RENEWALS constant)
- Extends due_date by the member tier loan_period_days

ERROR RESPONSES

Status	Message
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422	Maximum renewals reached.
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Return Loan

Bearer Token

<http://library-api.test/api/loans/1/return>

Staff token required (librarian or admin)

EXAMPLE RESPONSE

```
{
  "message": "Book returned successfully.",
  "data": {
    "id": 1,
    "status": "returned",
    "borrowed_at": "2026-06-22 14:50:54",
    "due_date": "2026-07-20 14:50:54",
    "returned_at": "2026-06-22 14:56:58",
    "renewals_count": 1,
    "fines_accrued": "0.00",
    "is_overdue": false
  }
}
```

BUSINESS RULES

- Staff only — librarian or admin
- If returned after due_date, a fine is automatically created
- Fine = days overdue x tier fine_rate (N50/day default)
- If total unpaid fines exceed N5,000 the member is auto-suspended

Reservations

POST

Create Reservation

Bearer Token

<http://library-api.test/api/reservations>

Member token required

REQUEST BODY

Creates a FIFO queue reservation

```
{"book_id": 2}
```

EXAMPLE RESPONSE

```
{
  "message": "Reservation created successfully.",
  "data": {
    "id": 1,
    "status": "pending",
    "queue_position": 1,
    "reserved_at": "2026-06-22 15:03:10",
    "notified_at": null,
    "claim_expires_at": null,
    "is_claim_expired": false
  }
}
```

BUSINESS RULES

- FIFO queue — queue_position assigned automatically
- Member has 48 hours to claim once notified (claim_expires_at)
- Cannot reserve a book already borrowed by the same member

GET

List Reservation

Bearer Token

<http://library-api.test/api/reservations>

Member token required

DELETE

Cancel Reservation

Bearer Token

<http://library-api.test/api/reservations/1>

Member token required

EXAMPLE RESPONSE

```
{"message": "Reservation cancelled successfully."}
```

Fines

GET

List Fines

Bearer Token

<http://library-api.test/api/fines>

Member token required

PATCH

Pay Fine

Bearer Token

<http://library-api.test/api/fines/1/pay>

Staff token required (librarian or admin)

Members

POST

Register Member

Public

<http://library-api.test/api/members>*No authentication required — public endpoint*

REQUEST BODY

```
{
  "first_name": "Chidi",
  "last_name": "Okeke",
  "email": "chidi@member.com",
  "password": "password",
  "password_confirmation": "password",
  "phone_number": "08011112222",
  "member_tier_id": 1
}
```

EXAMPLE RESPONSE

```
{
  "message": "Member registered successfully.",
  "data": {
    "id": 3,
    "first_name": "Chidi",
    "last_name": "Okeke",
    "email": "chidi@member.com"
  }
}
```

GET

List Members

Bearer Token

<http://library-api.test/api/members>*Staff token required (librarian or admin)*

GET

Get Single Member

Bearer Token

<http://library-api.test/api/members/1>*Staff token required (librarian or admin)*

PATCH

Update Member

Bearer Token

<http://library-api.test/api/members/1>*Member token required*

REQUEST BODY

Member can only update their own profile

```
{"phone_number": "08099999999"}
```

PATCH

Suspend Member

Bearer Token

<http://library-api.test/api/members/2/suspend>*Staff token required (librarian or admin)*

EXAMPLE RESPONSE

```
{"message": "Member suspended successfully.", "data": {"id": 2, "status": "suspended"}}
```

BUSINESS RULES

- Manual suspension by staff — independent of fine threshold

PATCH

Reinstate Member

Bearer Token

<http://library-api.test/api/members/2/reinstate>
Staff token required (librarian or admin)

EXAMPLE RESPONSE

```
{ "message": "Member reinstated successfully.", "data": { "id": 2, "status": "active" } }
```

BUSINESS RULES

- Restores member status to active

PATCH

Upgrade Member Tier

Bearer Token

<http://library-api.test/api/members/1/upgrade-tier>
Staff token required (librarian or admin)

REQUEST BODY

Staff only — admin/librarian

```
{ "member_tier_id": 2 }
```

BUSINESS RULES

- Regular: 3 books, 14 days | Premium: 6 books, 21 days

Member Tiers

GET

List Member Tiers

Public

<http://library-api.test/api/member-tiers>
No authentication required — public endpoint

POST

Create Tier

Bearer Token

<http://library-api.test/api/member-tiers>
Staff token required (librarian or admin)

REQUEST BODY

Admin only

```
{ "name": "Student", "max_books": 2, "loan_period_days": 7, "fine_rate": 50 }
```

PATCH

Update Tier

Bearer Token

<http://library-api.test/api/member-tiers/1>
Staff token required (librarian or admin)

REQUEST BODY

Admin only

```
{ "max_books": 4 }
```

Staff Management

GET

List Staff

Bearer Token

<http://library-api.test/api/staff>

Staff token required (librarian or admin)

POST

Create Staff

Bearer Token

<http://library-api.test/api/staff>

Staff token required (librarian or admin)

REQUEST BODY

Admin only — role: librarian or admin

```
{
  "name": "New Librarian",
  "email": "newlibrarian@library.com",
  "password": "password",
  "password_confirmation": "password",
  "role": "librarian"
}
```

PATCH

Update Staff

Bearer Token

<http://library-api.test/api/staff/3>

Staff token required (librarian or admin)

REQUEST BODY

Admin only

```
{"name": "Updated Librarian"}
```

DELETE

Delete Staff

Bearer Token

<http://library-api.test/api/staff/3>

Staff token required (librarian or admin)

Reports

GET

Dashboard

Bearer Token

<http://library-api.test/api/reports/dashboard>

Staff token required (librarian or admin)

GET

Overdue Loans

Bearer Token

<http://library-api.test/api/reports/overdue>

Staff token required (librarian or admin)

GET

Low Stock

Bearer Token

<http://library-api.test/api/reports/low-stock>

Staff token required (librarian or admin)

GET

Most Borrowed

Bearer Token

<http://library-api.test/api/reports/most-borrowed>

Staff token required (librarian or admin)

Business Rules Summary

Member Tiers

- Regular: max 3 books, 14-day loan period
- Premium: max 6 books, 21-day loan period

Fines

- Fine rate: N50 per day overdue (configurable per tier)
- Unpaid fines block new loans
- Total unpaid fines exceeding N5,000 auto-suspend the member

Loans

- Maximum 2 renewals per loan
- A member cannot borrow the same book twice simultaneously
- Only non-damaged, available copies are assigned

Reservations

- FIFO queue — first come, first served
- Member has 48 hours to claim once notified
- Unclaimed reservations expire automatically

Books

- Books with active loans cannot be deleted — only retired
- Retired books are hidden from the public catalog

Suspension

- Members can be manually suspended by staff
- Auto-suspension triggers at N5,000 unpaid fines
- Suspension must be manually lifted by staff via reinstate endpoint